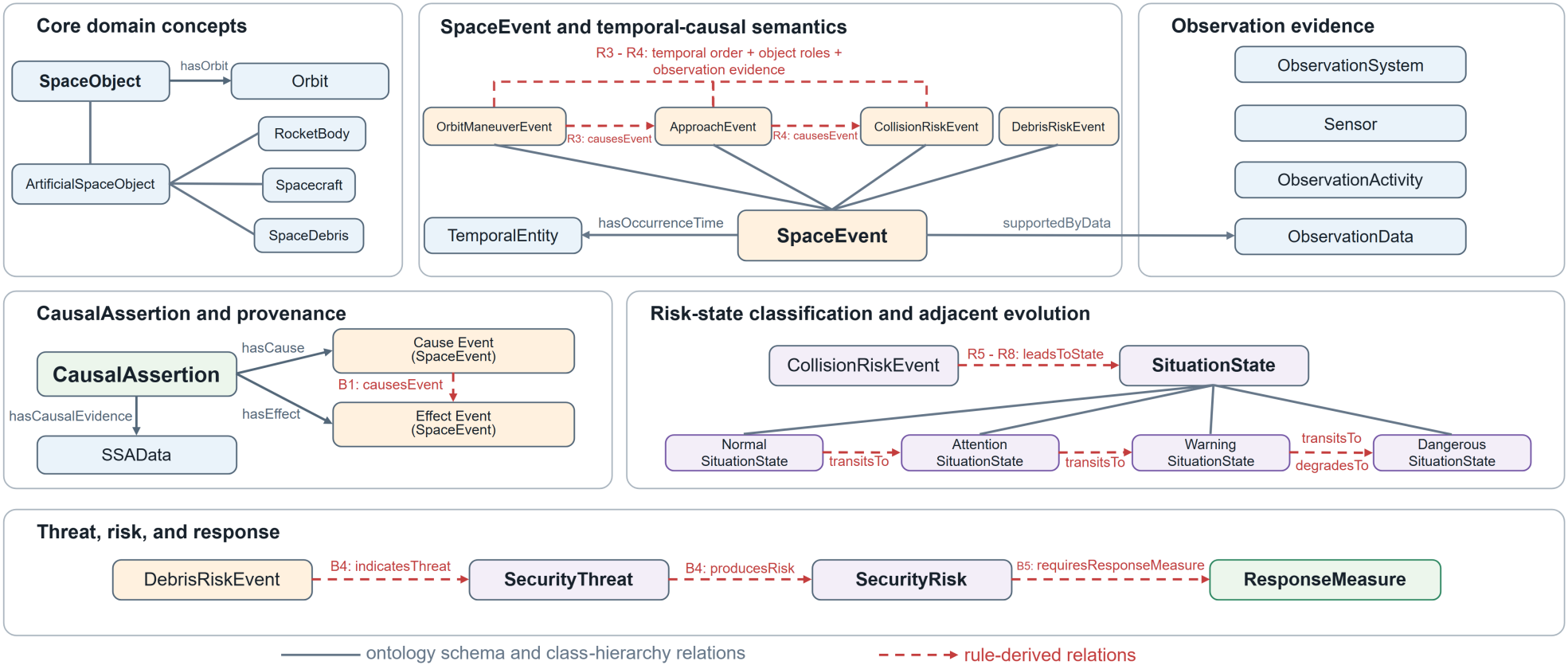

Core domain concepts
SpaceObject
hasOrbit
Orbit
RocketBody
ArtificialSpaceObject
Spacecraft
SpaceDebris
SpaceEvent and temporal-causal semantics
R3 - R4: temporal order + object roles + observation evidence
OrbitManeuverEvent
R3: causesEvent
ApproachEvent
R4: causesEvent
CollisionRiskEvent
DebrisRiskEvent
TemporalEntity
hasOccurrenceTime
SpaceEvent
supportedByData
Observation evidence
ObservationSystem
Sensor
ObservationActivity
ObservationData
CausalAssertion and provenance
CausalAssertion
hasCause
Cause Event (SpaceEvent)
B1: causesEvent
hasEffect
Effect Event (SpaceEvent)
hasCausalEvidence
SSAData
Risk-state classification and adjacent evolution
CollisionRiskEvent
R5 - R8: leadsToState
SituationState
Normal SituationState
transitsTo
Attention SituationState
transitsTo
Warning SituationState
transitsTo
degradesTo
Dangerous SituationState
Threat, risk, and response
DebrisRiskEvent
B4: indicatesThreat
SecurityThreat
B4: producesRisk
SecurityRisk
B5: requiresResponseMeasure
ResponseMeasure
ontology schema and class-hierarchy relations
rule-derived relations